

Lavanya Sunkara

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SUMMARY

AI/ML Engineer skilled in designing and deploying Agentic AI applications powered by LangChain, Hugging Face, and Groq. Experienced in end-to-end model integration, RAG pipelines, and scalable deployments across AWS, GCP, and Azure

AREAS OF EXPERTISE

- Programming Data: Python, SQL, C++ , Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- AI / ML Frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, TensorFlow, PyTorch, XGBoost, Random Forest, Decision Tree
- Agentic AI & LLM Tools: Groq Cloud (Llama-3.1-8B-Instant, gpt-oss models), GPT-4o-mini, RAG Systems, Autonomous Agent Workflows, Context-Aware Generation
- Backend & Web Development: FastAPI,Streamlit, Uvicorn,REST API,WebSocket,PDF-to-Vector Store Pipelines
- MLOps & Deployment: Docker,Kubernetes,Git,CI/CD, Jenkins,MLflow,Flask,FastAPI,REST APIs
- Databases & Storage: PostgreSQL,ChromaDB,Qdrant MySQL, MongoDB,PostgreSQL, VectorDB
- Automation & Data Handling: BeautifulSoup, Selenium,WebBaseLoader,Data Cleaning,Feature Engineering,Hyperparameter Tuning
- Computer Vision & AI Applications: OpenCV, TensorFlow Object Detection,Image Auto-Tagging
- AWS services - glue , lambda, redshift,Athena, IAM policies

PROFESSIONAL EXPERIENCE

AI/IT Intern, TiE New Jersey – Edison, NJ June 2025 – Aug 2025

- Designed AI-driven tools to enhance membership data access and system efficiency through backend and API integrations.
- Built a RAG chatbot using LlamaIndex and Qdrant, improving response relevance and reducing query latency.
- Developed a computer vision model (TensorFlow, OpenCV) to auto-tag event photos with 90 % detection accuracy.
- Automated data collection using Python, BeautifulSoup, and Selenium, increasing membership data coverage by 40%.
- Collaborated remotely with stakeholders to align AI solutions with organizational goals

Data Scientist , TATA Consultancy Services – India Aug 2019 – Dec 2023

- Used Python with Pandas and SQL on PostgreSQL to prepare sample datasets for digital and cloud demos, enabling smooth testing for 3,000 more employees capacity planning.
- Applied supervised learning in Python and XGBoost for basic predictive analytics in manufacturing, like energy forecasting, to aid LTTS's digital twin prototypes.
- Conducted exploratory data analysis (EDA) with NumPy and Matplotlib on enterprise data via the ISB MoU, identifying key digitization gaps for client reports.
- Built regression models using R and statistical testing to analyze energy use from IoT sensors, supporting i-BEMS rollout for 3% savings in plant trials.
- Collaborated with data engineers on AWS EC2 and Spark for ETL processes, merging LTI datasets into LTIMindtree systems without downtime.
- Performed clustering analysis on public safety data with Python to prototype surveillance patterns, informing SWC acquisition strategies for fiber grid connectivity.
- Used Tableau for visualizing urban mobility trends from metro data, sharing findings in Scrum meetings to guide feeder service enhancements.

- Optimized model performance and feature pipelines using Scikit-learn and MLflow, streamlining experimentation tracking, and improving model reproducibility across multiple analytics projects.

PROJECTS

Cold Email Generator (AI-Powered Streamlit App)

github

- Built an AI app that reads resumes and generates personalized recruiter emails from job descriptions.
- Integrated Groq Cloud's Llama-3.1-8B-Instant via LangChain to automate context-aware email generation.
- Implemented web scraping and parsing using WebBaseLoader and PyPDF2 to extract key role details.
- Deployed an interactive Streamlit interface for seamless user input and instant email generation.

Coder Buddy – Agentic AI Application Generator

github

- Built an end-to-end AI “coder buddy” that converts natural language prompts (e.g., “create a calculator web app”) into fully functional applications
- Designed autonomous agent workflows with LangChain and LangGraph to plan tasks, generate code, and assemble complete projects.
- Used gpt-oss models on Groq Cloud for low-latency, multi-file code generation across frontend, backend, and project setup.
- Developed modular agents (planner, architect, executor) to simulate the software development lifecycle and boost developer productivity.
- Reduced prototype setup time from hours to minutes, showcasing the potential of AI-augmented software engineering.

Agentic Ecommerce Assistant

github

- Built a production-ready AI assistant for ecommerce using GPT-4o-mini with LangGraph, enabling natural language catalog queries, cart management, promotions, and order tracking.
- Designed a retrieval-augmented search (RAG) system that connects product data (PostgreSQL + Chroma embeddings) and store policy FAQs (PDF → vector store) to ensure accurate, grounded responses.
- Delivered a FastAPI + Uvicorn backend with REST and streaming WebSocket APIs, paired with a lightweight WebSocket chat UI and PostgreSQL persistence for catalog and conversation history.
- Implemented session-aware cart logic, promotion handling, order status lookups, and chat archiving; automated onboarding with database seeding and vector indexing.
- Tech Stack: GPT-4o-mini, LangGraph, FastAPI, Uvicorn, PostgreSQL, ChromaDB, WebSocket, REST API, PDF-to-Vector Store, Git/GitHub

NYC Property Price Prediction

- Analyzed over 77,000 housing records to clean, standardize, and prepare data for predictive modeling.
- Built a complete machine learning pipeline and improved model accuracy by resolving data inconsistencies and outliers.
- Compared multiple regression models and achieved $R^2 = 0.99$ with XGBoost using 5-fold cross-validation.
- Tuned model parameters to enhance reliability and deliver highly accurate property price predictions.
- Tech Stack: Python, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, Decision Tree, Matplotlib, Seaborn

EDUCATION

New Jersey Institute of Technology | Newark, NJ, MS in Computer Science

Jan 2024 – Aug 2025

- GPA: 3.65/4.0

Bapatla Engineering College | Bapatla, India, Bachelor of Technology in
Computer Science & Engineering

Aug 2015 – May 2019

- GPA: 8.1/10.0